CHEMISTRY STRATEGIC TOPIC ANALYSIS

Based on PYQ Sets: RP5PS/5, WYXZ7, and HFG1E/5

1. PHYSICAL CHEMISTRY

Chemical Kinetics (Rate Laws & Order)

- **Integrated Rate Equations:** Constant appearance of numericals comparing *t_{99%}* and *t_{90%}* for first-order reactions.
- **Arrhenius Equation:** Mandatory prep for relations between temperature, rate constants (k), and Activation Energy (Ea).
- **Order of Reaction:** Frequent checks on determining order from rate data or rate constant units.

Electrochemistry (Nernst & Conductivity)

- **Nernst Equation:** Guarantee on EMF (Ecell) and Gibbs Free Energy calculation.
- **Conductivity:** Kohlrausch law and the variation of molar conductivity with concentration is a high-frequency reasoning area.
- **Faraday's Laws:** Quantitative electrolysis numericals.

Solutions (Colligative Properties)

- **Boiling/Freezing Points:** Numericals on Delta Tb and Delta Tf. Must include **Van't Hoff factor (i)** for salts like MgCl2.
- **Raoult's Law:** Identification of positive/negative deviations from ideal behavior.

2. INORGANIC CHEMISTRY

Coordination Compounds (VBT & CFT)

- **VBT:** Prediction of hybridization (sp3, dsp2, d2sp3) and magnetic behavior for Ni/Co complexes.
- **CFT:** Explaining color and d-orbital splitting.
- **IUPAC & Isomerism:** Guarantee on Linkage, Geometrical, and Optical isomerism identification.

d- and f-Block Elements

- **Lanthanoid Contraction:** Its cause and consequences (Zr/Hf radii) are staples.
- **Trends in 3d series:** Reasoning on E° values and why certain oxidation states (like Mn3+/Cr2+) are reducing/oxidizing agents.

- **KMnO4 & K2Cr2O7:** Ionic equations and preparation methods.

3. ORGANIC CHEMISTRY

Mechanisms & reactivity

- **SN1 vs SN2:** Ranking reactivity of haloalkanes based on steric hindrance and carbocation stability.
- **Nucleophilic Addition:** Aldehydes vs Ketones reactivity orders.

Name Reactions & Conversions

- **Recurring:** Cannizzaro, Aldol, Clemmensen, Sandmeyer, and Etard reactions.

Acidity, Basicity & Chemical Tests

- **Distinction Tests:** Iodoform, Tollens', Fehling's, and Hinsberg Test.
- **Acidity/Basicity:** Phenol acidity (effect of -NO2) and Aliphatic Amine basicity orders in aqueous phase.

4. BIOMOLECULES

- **Glucose Structure:** Chemical evidence for CHO group, 5-OH groups, and straight chain.
- **Proteins/Vitamins:** Peptide linkage, Denaturation, and Vitamin deficiency diseases.